Fine-tuning Geneformer for Therapeutic Target Discovery in COVID-19

In Silico Knockout Analysis on Single-Cell RNA-seq Data

Manuella Kristeva NAKAM YOPDUP

manuella.yopdup@aims.ac.rw
Zuse Institute Berlin(ZIB)
Berlin
Germany

BNITM
Bernhard-Nocht-Institut für Tropenmedizin

About Me

Manuella Kristeva NAKAM YOPDUP

- Combine background in Computer Science and Applied Mathematic
- Currently working on the pruning of Transformers that could be transferable to the project
- Knowledges in programming, linux environments and HPC
- Passionate by AI applied to Health then motivated to learn
- I have differents experience in research
- train, explore and modify transformer-based models, especially
- Contribute to transfer learning and predictive pipelines
- I am happy to provide bioinformatics support to other groups and enjoy collaborations
- Adaptative and quick learner

1. About me
2. Geneformer

Task Objective

The goal of this project was to :

- Fine-tune Geneformer on a public infection-related scRNA-seq dataset
- Focus on one immune cell population
- Perform in silico gene knockout analysis
- Interpret the biological results
- Critically evaluate the strengths and limitations of the approach

How does Geneformer work ? ? [1]

- Input : single-cell RNA-seq matrix where "rows = cells" and "columns = genes"
- Geneformer ranks genes by expression within each cell
- Tokenization : Each gene has a unique token ID
- The model uses Self-Attention to learn which genes "attend" to each other
- The model was pre-trained on 30 million cells from the Genentech/Theodoris dataset
- After pretraining, the model can be adapted for downstream tasks(cell classification, disease prediction,..)
- In silico perturbation : Geneformer can simulate gene deletion to measure how the cell state changes

Dataset Selection

- Covid 19 : it is an infectious disease caused by the SARS-CoV-2 virus.
- "Single-cell atlas of peripheral immune response to SARS-CoV-2 infection" dataset from CELLxGENE.
- ".h5ad" format with around 44000 cells and around 25000 genes
- Available metadata (disease, cell_type, donor_id, sex, tissue, development stage)
- Selected cell population : CD14-positive monocytes
  1. In severe COVID-19, $CD14^+$ monocytes show high levels of NLRP3 inflammasome activation, leading to pyroptosis (inflammatory cell death)
  2. $CD14^+$ monocytes in infected patients express elevated levels of inflammatory markers and cytokines
- Monocytes are highly inflammatory, strongly responsive to viral infection, rich in interferon signaling.
- For the experiments we are using 1000 COVID monocytes and 1000 healthy monocytes (to avoid classification bias)

Preprocessing

- Load .h5ad dataset with Scanpy
- Filter $CD14^{+}$ Monocytes to eliminate noise from other immune populations (T-cells, B-cells).
- Restrict to "Normal" and "COVID-19" to create a binary classification task.
- Subsample to obtain 1,000 cells per class.
- Creates an explicit column for Ensembl IDs because Geneformer's vocabulary is built on Ensembl IDs
- Calculates the total number of transcript counts per cell to perform the rank-value encoding accurately

Tokenization

- Convert each single-cell transcriptomes into transformer-compatible token sequences.
- Defines which metadata (cell type, donor, disease) should be saved alongside the gene data.
- Initializes the tokenizer and set nproc=4 uses multi-core processing to speed up the ranking of 2,000 cells.
- Ranks the genes in every cell by expression and keeps the top 2,048. It converts these ranks into tokens (numbers) that represent the genes.

Example

{ 'input_ids' : [2, 18332, 5029, 9515, 1616, 2925, 15985, ...],
'cell_type' : 'CD14-positive monocyte',
'disease' : 'COVID-19',
'donor_id' : 'C4',
'length' : 1349
}

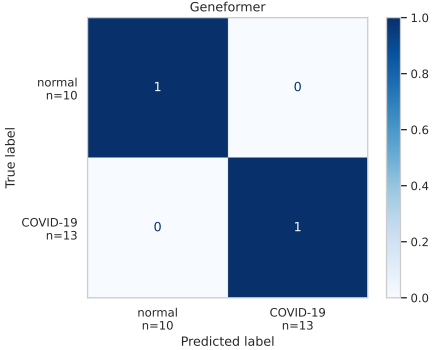
Fine-Tuning Setup : Binary Classification

- Model Used : Geneformer V2-104M (Good compromise between biological representation quality, runtime, GPU memory)
- Configuration : freeze_layers = 2, forward_batch_size = 50, one cross-validation split
- 90% Training / 10% Testing

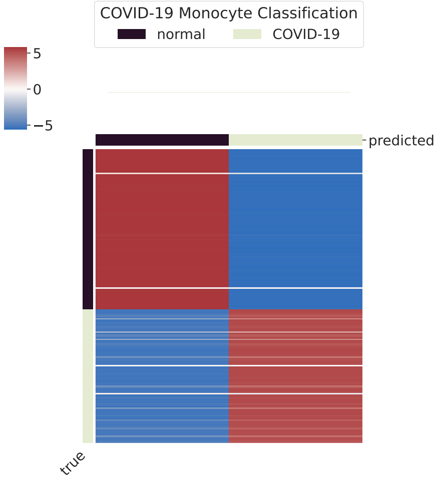
Table – Model Test Evaluation Metrics

Metric	**Value**
Test Accuracy	0.9900
Test Loss	0.0147
Test Macro F1	0.9899
Test Runtime (seconds)	6.3783
Test Samples per Second	31.3560
Test Steps per Second	2.6650

Results Fine-Tuning



Confusion Matrix(last batch)



The Prediction Heatmap

In Silico Perturbation Analysis

- Identify genes that, when deleted, shift the COVID-19 cell state toward the Normal state.

Table – Gene Analysis Metrics

Gene Name	Ensembl ID	Shift to Goal	P-val	FDR	N Detections
DPM1	ENSG00000000419	0.0	1.0	1.0	4
FIRRM	ENSG00000000460	0.0	1.0	1.0	1
FGR	ENSG00000000938	0.0	1.0	1.0	56
FUCA2	ENSG00000001036	0.0	1.0	1.0	14
GCLC	ENSG00000001084	0.0	1.0	1.0	5
NFYA	ENSG00000001167	0.0	1.0	1.0	2
STPG1	ENSG00000001460	0.0	1.0	1.0	1
NIPAL3	ENSG00000001461	0.0	1.0	1.0	5

Why Were the Perturbation Results Weak ?

- Small dataset size ($\sim$2000 cells), making perturbation estimates less stable.
- Very small evaluation set, which may limit the robustness of the learned disease separation.
- Geneformer is a large model and may require more data or tuning for stable perturbation analysis.
- Perturbation is performed in embedding space and does not directly model biological gene regulation.
- Some genes were detected in very few cells, reducing perturbation reliability.
- COVID-19 immune responses are driven by many coordinated genes, so deleting one gene may have limited impact.

Biological Insights From the Experiment

- This suggests that COVID-19 induces major transcriptional remodeling in CD14-positive monocytes.
- CD14-positive monocytes are highly responsive during viral infection (COVID-19)
- No single gene strongly controlled the disease-state transition.
- Weak perturbation shifts may reflect biological robustness
- Deleting one gene computationally may not strongly alter because many related genes compensate.

Conclusion

Classification

Question : Can the model distinguish disease states globally ?
Answer : Yes

In silico pertubations

Question : Does one gene strongly control the latent disease state ?
Answer : Apparently not strongly in the chosen dataset

Implementation Decisions and Challenges

- Addressed a known mismatch between Geneformer and newer Hugging Face transformers versions.
- System utilized Python 3.13 ; Geneformer requires Python 3.11.
- Forced PYTHONNOUSERSITE=1 and PYTHONPATH injection to prevent "library bleeding" from local user directories
- Injected a MockTokenizer into the Data Collator to bypass AttributeError in the Trainer class.
- Patched the InSilicoPerturberStats module to handle integer labels (0,1) instead of strings, preventing TypeError during ranking.
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Bibliographie

C. V. Theodoris, L. Xiao, A. Chopra, M. D. Chaffin, Z. R. Al Sayed, M. C. Hill, H. Mantineo, E. M. Brydon, Z. Zeng, X. S. Liu & P. T. Ellinor – « Transfer learning enables predictions in network biology », *Nature* **618** (2023), no. 7965, p. 616–624.